

## §12.6 The chain rule

$$\text{If } y = f(x(t)) \text{ then } \frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt}$$

$$\begin{aligned} \text{eg } y &= e^{\sin x} \\ \text{let } x &= \sin t \end{aligned} \quad \frac{dy}{dt} = \frac{d e^x}{dx} \frac{dx}{dt} = e^x \cos t = e^{\sin t} \cos t$$

$$\text{If } z = f(x, y) \text{ with } x = x(t), y = y(t) \text{ then}$$

$$\boxed{\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}} = D_z \cdot (x', y')$$

$$\begin{aligned} \text{Proof } x(t+h) &= x(t) + x'(t) \cdot h + \epsilon_1(h) \cdot h & \lim_{h \rightarrow 0} \epsilon_i(h) &= 0 \\ y'(t+h) &= y'(t) + y'(t) \cdot h + \epsilon_2(h) \cdot h \end{aligned}$$

$$f(x(t+h), y(t+h)) = f(x + \Delta x, y + \Delta y) = f(x, y) + f_x \cdot \Delta x + f_y \cdot \Delta y + \epsilon_x \cdot \Delta x + \epsilon_y \cdot \Delta y$$

$$\begin{aligned} &= f(x, y) + f_x \cdot (x'(t) \cdot h + \epsilon_1(h) \cdot h) \\ &\quad + f_y \cdot (y'(t) \cdot h + \epsilon_2(h) \cdot h) \\ &\quad + \epsilon_x (x'(t) \cdot h + \epsilon_1(h) \cdot h) + \epsilon_y (y'(t) \cdot h + \epsilon_2(h) \cdot h) \end{aligned}$$

$$\frac{f(x(t+h), y(t+h)) - f(x, y)}{h} = f_x' \cdot x' + f_y \cdot y' + \underbrace{\left( \epsilon_1(h) + \epsilon_2(h) + \epsilon_x \cdot (x' + \epsilon_1) + \epsilon_y (y' + \epsilon_2) \right)}_{\downarrow 0}$$

We have assumed that  
 $f(x, y)$  is differentiable and  
so are  $x(t), y(t)$ .

$$\text{Eg } f(x, y) = x^2/y^2 \quad x = t^2 - 1 \quad y = t^3 + 1$$

$$\begin{aligned} \frac{df}{dt} &= \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} = \frac{2x}{y^2} \cdot 2t + \frac{2x^2}{y^3} \cdot 3t^2 \\ &= \frac{2(t^2-1)}{(t^3+1)^2} \cdot 2t + \frac{2(t^2-1)^2}{(t^3+1)^3} \cdot 3t^2 \end{aligned}$$

$$\text{Or } f(t) = \frac{(t^2-1)^2}{(t^3+1)^2}$$

$$f'(t) = \frac{2(t^2-1) \cdot 2t}{(t^3+1)^2} + \frac{2(t^2-1)^2 \cdot 3t^2}{(t^3+1)^3}$$

Eg The volume of a cylinder is  $V = 2\pi r h$

If  $r = 10\text{m}$ ,  $h = 10\text{m}$  and  $r$  is increasing by  $.1\text{m/s}$  while  $h$  is decreasing by  $-.2\text{m/s}$ .

How fast is the volume of the cylinder changing by?

$$\begin{aligned} \frac{dV}{dt} &= \frac{\partial V}{\partial r} \frac{dr}{dt} + \frac{\partial V}{\partial h} \frac{dh}{dt} = 2\pi h \cdot .1 + 2\pi r \cdot (-.2) \\ &= 2\pi(10) \cdot .1 - 2\pi(10) \cdot (-.2) = -2\pi \end{aligned}$$

Suppose  $x = x(s, t)$   $y = y(s, t)$   $z = f(x, y)$   
are differentiable, then

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$

"Better"

$$\begin{pmatrix} z_s \\ z_t \end{pmatrix} = \begin{pmatrix} x_s & y_s \\ x_t & y_t \end{pmatrix} \begin{pmatrix} z_x \\ z_y \end{pmatrix}$$

$$D_{s,t} z = D_{s,t}(x, y) \cdot D_{x,y}(z)$$

eg.  $w = x^2 y - x y^2$   $x = s+t$   $y = s+t+1$

$$\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial t}$$

$$= (2xy - y^2)s + (x^2 - 2xy) = (2s + (s+t) - (s+t)^2)s + ((s+t)^2 - 2s(s+t))$$

eg  $w = e^{xy}$

$$x = \sin(s+t) \quad y = t$$

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial s} = y e^{xy} \cos(s+t) + x e^{xy} \cdot 1$$

$$= t e^{t \sin(s+t)} \cos(s+t) + \sin(s+t) e^{t \sin(s+t)}$$

# Implicit differentiation

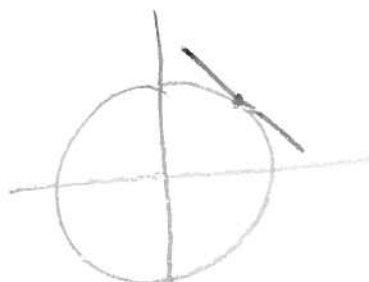
$$F(x, y) = 0$$

$$\text{eg } x^2 + y^2 - 2 = 0$$

$$\frac{\partial F}{\partial x} \frac{dx}{dx} + \frac{\partial F}{\partial y} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = - \frac{\partial F / \partial x}{\partial F / \partial y}$$

$$P = (1, 1) \quad \frac{dy}{dx} = \frac{-2x}{2y} = -1$$



$$\text{Eg } y^2 - xy + x^3 - 1 = 0$$

at (1, 1)

$$\frac{dy}{dx} = \frac{-(-y + 3x^2)}{2y - x} = \frac{-2}{1} = -2$$

Ex 1 Find  $\frac{dw}{dt}$  using the chain rule. Express your answer in terms of  $t$ .

a)  $w = x^2 y^3$ ,  $x = t^3$ ,  $y = t^2$

b)  $w = e^x \sin y + e^y \sin x$ ,  $x = 3t$ ,  $y = 2t$

c)  $w = \sin(xyz^2)$ ,  $x = t^3$ ,  $y = t^2$ ,  $z = t$

Ex 2 Find  $\frac{\partial w}{\partial t}$  using the chain rule. Express answer in terms of  $s, t$ .

a)  $w = x^2 y$ ,  $x = st$ ,  $y = s - t$

b)  $w = e^{x^2 + y^2}$ ,  $x = s \sin t$ ,  $y = t \sin t$

c)  $w = x^2 y$ ,  $x = 2t + s$ ,  $y = 1 - st$ . Find  $\frac{\partial w}{\partial t} \big|_{s=1, t=-2}$

Ex 3 Find  $dy/dx$  by implicit differentiation

a)  $x^3 + 2x^2 y - y^3 = 0$

b)  $x \sin y + y \cos x$